
On the Multiplicity of Predictions in Classification

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Abstract

In the context of machine learning, a prediction problem exhibits *multiplicity* if there exist several “good” models that attain identical or near-identical performance (i.e., accuracy, AUC, etc.). In this paper, we study the effects of multiplicity in human-centric applications, such as credit scoring and recidivism prediction. We introduce a specific notion of multiplicity – *predictive multiplicity* – to describe the existence of good models with conflicting predictions. We propose measures to evaluate predictive multiplicity in classification problems. We present integer programming methods to compute these measures, and show that recidivism prediction problems often admit many good models that output wildly conflicting predictions.

1 Introduction

Many methods in machine learning are designed to fit the *best* predictive model. If stakeholders agree on the formulation of the prediction problem [13, 17], and we have the tools to solve it, there is still a key issue with this approach: *there may be more than one “best” model*. In machine learning, the notion that a prediction problem may not have one “best” model but many “good” models that attain near-identical performance is called *multiplicity* [see 3, 12, 14].

In *human-centric applications*, such as credit scoring and recidivism prediction, the existence of near-optimal models with conflicting predictions challenges key aspects of how we build, interact with, and trust models. Ignoring the existence of competing models would mean that we would arbitrate conflicting predictions by brute luck. In this paper, we introduce a general framework to evaluate the multiplicity of predictions in classification problems. The main contributions of our paper are:

1. We present formal measures of how multiplicity affects individual predictions.
2. We develop integer programming tools to compute these measures for linear classification problems in a way that is exact and efficient. This is non-trivial since computing these measures requires solving hard optimization problems over a user-defined set of good models.
3. We present an empirical study of predictive multiplicity in recidivism prediction. Our results show that many real-world datasets can admit good models with wildly conflicting predictions.
4. We provide a software implementation of our tools in Python at [url redacted].

(x_1, x_2)	# EXAMPLES		OPTIMAL CLASSIFIERS					
	n^+	n^-	$\hat{h}_0$	$\hat{h}_a$	$\hat{h}_b$	$\hat{h}_c$		
(0, 0)	0	100	—	—	—	+	Training Error of Optimal Classifier	25%
(0, 1)	100	0	+	—	+	+	Test Error of Optimal Classifier	25%
(1, 0)	100	0	+	+	—	+	Discrepancy among Optimal Classifiers	50%
(1, 1)	0	100	+	—	—	—	Ambiguity among Optimal Classifiers	100%

Figure 1: Toy example showing how multiple linear classifiers can perform equally well while assigning conflicting predictions. Here, n^+ and n^- denote the number of examples with $y = +1$ and $y = -1$ for each $(x_1, x_2) \in \{0, 1\}^2$, $\hat{h}_0, \hat{h}_a, \hat{h}_b, \hat{h}_c$ represent linear classifiers that minimize training and test error (25%). The predictions of any 2 classifiers differ on 200 points (50% of the training sample). Moreover, there exists an optimal classifier that can predict +1 or -1 for each point.

1.1 Related Work

Recent work in machine learning tackles multiplicity from alternative perspectives. Fisher et al. [7] and Dong and Rudin [6] present methods to evaluate variable importance over the set of good models. Semenova and Rudin [19] propose a measure for the relative size of the set of good models, and use it to characterize prediction problems where simple models perform well. Our work differs from this stream of research in that we focus on problems where good models output conflicting predictions (see e.g., Figure 1). This type of multiplicity leads to irreconcilable trade-offs between subsets of individuals, which is similar to the impossibility results in the fair machine learning literature [e.g., 4, 9, 5, 10].

2 Framework

2.1 Preliminaries

We are given a training dataset of n examples $\{(x_i, y_i)\}_{i=1}^n$ where each *example* consists of a vector of $d + 1$ *features* $x_i = (1, x_{i1}, \dots, x_{id}) \in \mathbb{R}^{d+1}$ and a *label* $y_i \in \{-1, +1\}$. Our goal is to train a classifier $h : \mathbb{R}^{d+1} \rightarrow \{-1, +1\}$ from a fixed hypothesis class $\mathcal{H}$ that optimizes the true risk (i.e., test error). To this end, we minimize the empirical risk (i.e., the training error): $\hat{R}(h) = \sum_{i=1}^n \mathbb{1}[h(x_i) \neq y_i]$. We denote an optimal classifier as $\hat{h} \in \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}(h)$. Unless otherwise specified, we will assume that we deploy a baseline classifier $\hat{h}_0 = \hat{h}$ that optimizes performance on training data.¹ Given a baseline classifier $\hat{h}_0$ and an error tolerance ϵ , we refer to the set of classifiers whose performance differs by at most ϵ as the ϵ -level set, denoted $S_\epsilon(\hat{h}_0)$. We say a prediction problem exhibits *multiplicity* if the ϵ -level set around $\hat{h}_0$ is non-trivial (i.e., if $|S_\epsilon(\hat{h}_0)| > 1$). We should set ϵ so that the ϵ -level set will, with high probability, contain the model that optimizes performance in deployment.²

2.2 Predictive Multiplicity

A prediction problem exhibits *predictive multiplicity* if there exist several models that attain similar performance but that produce conflicting predictions on the training data.

Definition 1 (Predictive Multiplicity) *Given a baseline classifier $\hat{h}_0$ and an error tolerance ϵ , a prediction problem exhibits predictive multiplicity over the ϵ -level set $S_\epsilon(\hat{h}_0)$ if there exists a model $h \in S_\epsilon(\hat{h}_0)$ such that $h(x_i) \neq \hat{h}_0(x_i)$ for some x_i in the training dataset.*

Any incidence of predictive multiplicity implies that there is *concrete evidence* of conflicting predictions among the set of good models. Thus, we will have to choose between models that produce irreconcilable predictions on the training data. Given a baseline classifier $\hat{h}_0$, we measure the severity

¹Our framework and tools also apply for baseline models $\hat{h}_0$ that do not optimize training performance.

²In general, the error tolerance ϵ should be non-zero and can be determined by generalization bounds, or by computing confidence intervals of out-of-sample performance (e.g., via bootstrapping or cross-validation).

of predictive multiplicity by counting the number of conflicting predictions between $\hat{h}_0$ and a model in the ϵ -level set.

Definition 2 (Discrepancy) *The discrepancy of a baseline classifier $\hat{h}_0$ over the ϵ -level set $S_\epsilon(\hat{h}_0)$ is the maximum number of conflicts over points in a training set $\mathbf{x}_i \in X$ between $\hat{h}_0$ and any competing classifier $g_\epsilon \in S_\epsilon(\hat{h}_0)$. Formally, the discrepancy $\delta_\epsilon(\hat{h}_0)$ is defined:*

$$\delta_\epsilon(\hat{h}_0) := \max_{h' \in S_\epsilon(\hat{h}_0)} \#\{\mathbf{x}_i \in X : h'(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)\}.$$

If the discrepancy is large, there exists a near-optimal solution which makes very different predictions than the baseline classifier. This means practitioners should consider choosing between models on the basis of secondary objectives, and rule out the possibility of adversarial model selection.

If there exists a classifier in the ϵ -level set making a different prediction on $\mathbf{x}_i$, then the prediction $\hat{h}_0(\mathbf{x}_i)$ is susceptible to model choice. In this case, we say that the prediction on $\mathbf{x}_i$ is *ambiguous*. Individual ambiguity may provide grounds for an individual to contest their prediction.

Definition 3 (Ambiguity) *The ambiguity of a baseline classifier $\hat{h}_0$ over the ϵ -level set $S_\epsilon(\hat{h}_0)$ is the number of points in a training dataset $\mathbf{x}_i \in X$ that can be assigned a conflicting prediction by a competing classifier $g_i \in S_\epsilon(\hat{h}_0)$:*

$$\alpha_\epsilon(\hat{h}_0) := \#\{\mathbf{x}_i \in X : \exists h' \in S_\epsilon(\hat{h}_0) \text{ s.t. } h'(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)\}.$$

While discrepancy reflects the maximum number of predictions that can change between the baseline model and a single model in the ϵ -level set, ambiguity measures how many individual predictions are determined by the choice of one model between a set of good models. Considering the fact that these measures depend on our choice of baseline model, we view the ambiguity and discrepancy of an optimal model as the canonical measures of predictive multiplicity for a given prediction problem, and denote these quantities as α_ϵ and δ_ϵ , respectively. See our extended paper [citation redacted] for formal guarantees on the discrepancy and the number of good models.

3 Methodology

In this section, we present integer programming tools to compute our measures of multiplicity for linear classification problems. We express each optimization task as a *mixed integer program* (MIP).

Training the Baseline Classifier We train an optimal classifier $\hat{h}$ that directly minimizes the training error (i.e., the 0-1 loss function) by solving an optimization problem of the form: $\min_{h \in \mathcal{H}} \sum_{i=1}^n \mathbb{1}[h(\mathbf{x}_i) \neq y_i]$. Due to space constraints, we refer the reader to our extended paper [citation redacted] for the MIP formulation we use to solve this optimization problem.

Computing Discrepancy Given a training dataset, a baseline classifier $\hat{h}_0$, and a specific error tolerance ϵ , we compute the maximum discrepancy among all models within the ϵ -level set of $\hat{h}_0$ by solving the optimization problem $\min_{h \in \mathcal{H}} \sum_{i=1}^n \mathbb{1}[h(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)]$ such that $|\hat{R}(h) - \hat{R}(\hat{h}_0)| \leq \epsilon$. The solution to this optimization problem is a classifier in the ϵ -level set that maximizes discrepancy with respect to the baseline classifier, which we denote as g_ϵ . See our extended paper [citation redacted] for the MIP formulation we propose to solve this optimization problem.

Computing Ambiguity Given a training dataset and a baseline classifier $\hat{h}_0$, we compute the value of ambiguity by training a pathological classifier g_i for each point $i = \{1, \dots, n\}$. Given the baseline classifier $\hat{h}_0$ and the instance $\mathbf{x}_i \in X$, the pathological classifier for point i g_i is the most accurate linear classifier that outputs a conflicting prediction on point i . We can train g_i by solving an optimization problem of the form $\min_{h \in \mathcal{H}} \sum_{k=1}^n \mathbb{1}[h(\mathbf{x}_k) \neq y_k]$ such that $h(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)$. See our extended paper [citation redacted] for the MIP formulation we propose to solve this optimization problem. Given a set of pathological classifiers g_i for each point $i \in \{1, \dots, n\}$, we can recover the ambiguity over the ϵ -level set as the number of pathological classifiers whose performance is within ϵ of the baseline model by computing $\alpha_\epsilon = \#\{\mathbf{x}_i \in X : \hat{R}(g_i) \leq \hat{R}(\hat{h}_0) + \epsilon\}$.

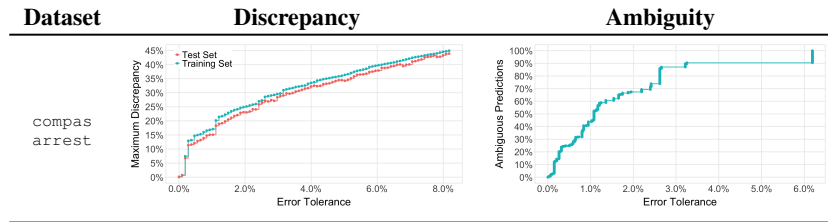


Figure 2: Overview of discrepancy (left) and ambiguity (right) computed over the ϵ -level set (i.e., the set of models whose training error is within the error tolerance of the optimal training error) Our analysis for `compas_arrest` shows that the set of models for $\epsilon = 1\%$ exhibit a discrepancy of 17%. This means that one can change 17% of predictions by switching the baseline model with a model that is only 1% less accurate. Ambiguity reflects the fact that 44% of predictions are susceptible to model selection within this set. We observe that the values of discrepancy and ambiguity are similar on both the training and test set. We include results for other datasets in Appendix C.

4 Experiments

In this section, we present experiments where we apply our tools to evaluate predictive multiplicity in recidivism prediction problems.³ We define a set of 9 recidivism prediction problems from 3 studies of recidivism in the United States: `pretrial` [16], `recidivism` [15], and `compas` [1]. We present an overview of each dataset and the processing details in Appendix C. For each dataset, we train a baseline classifier and competing classifiers to measure discrepancy and ambiguity, as described in Section 3. We show results for the `compas_arrest` dataset in Figure 2 and Table 2. See Appendix C for the full experimental results.

4.1 Results

Our results across all 9 datasets show that between 4% and 53% of individuals are assigned ambiguous predictions by models in the 1%-level set. In `compas_arrest`, for example, we observe an ambiguity of 44%. This means that 2,355 individuals (44%) are assigned different predictions by a competing model. For `compas_violent`, ambiguity over the 1%-level set is 53%, which means *the majority of predictions are affected by which competing model is chosen for deployment*. Considering discrepancy, we see that for `compas_arrest` we can find a *single competing model* in the 1% level set that assigns 17% of individuals a different prediction.

We also find that the incidence of multiplicity can differ significantly between subgroups. In `compas_violent`, for example, the proportion of individuals whose predictions are ambiguous within the 1% level set is: 72.9% among African-American individuals ($n = 4,042$), 37.2% among Caucasian individuals ($n = 3,923$), 32.5% among Hispanic individuals ($n = 579$), and 34.4% among Other individuals ($n = 224$). In this case, we find that multiplicity disproportionately affects African-Americans compared to individuals of other ethnic groups. Groups with a larger burden of multiplicity are more vulnerable to model selection, and more likely to be affected by the ignorance of competing models.

5 Concluding Remarks

Predictive multiplicity can arise in many real-world problems, and has important implications for how we choose and interact with predictive models. In applications such as recidivism prediction and credit scoring, predictive multiplicity implies irreconcilable differences in individual predictions. Thus, when a prediction problem exhibits severe predictive multiplicity, we need to carefully justify our decision to choose one model over other models that perform equally well. If there exists different models that produce conflicting predictions on individuals, and no justification to support one model over another, then it may be a better choice to forgo deployment entirely, or to assign favorable predictions to individuals who receive ambiguous predictions.

³We consider recidivism prediction because it is a domain where the existence of predictive multiplicity has serious ethical implications and may present an additional critique [see e.g., 8, 11, 2, for a critique].

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A Depiction of Types of Multiplicity

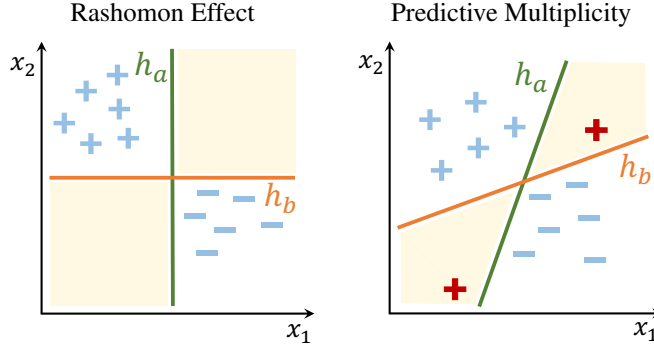


Figure 3: The “Multiplicity of Good Models” does not necessarily imply a “Multiplicity in Prediction.” We compare classification problems where we can find two classifiers that optimize accuracy: h_a and h_b . We highlight regions of conflict in yellow, and points that are assigned conflicted predictions in red. On the left, h_a and h_b assign the same predictions on the training data (but produce conflicting explanations of the importance of x_1 vs. x_2 , as per the Rashomon effect). On the right, h_a and h_b assign conflicting predictions on the training data (i.e., predictive multiplicity). A prediction problem can exhibit both effects, but predictive multiplicity requires that there is concrete evidence of irreconcilable differences between the predictions of good models.

B Datasets and Preprocessing

Dataset	n	d	Outcome Variable
compas_arrest	5,380	18	rearrest for any crime
compas_violent	8,768	18	rearrest for violent crime
pretrial_CA_arrest	9,926	22	rearrest for any crime
pretrial_CA_fta	8,738	22	failure to appear
pretrial_NY_fta	730	22	failure to appear
recidivism_CA_arrest	114,522	20	rearrest for any crime
recidivism_CA_drug	96,664	20	rearrest on drug crime
recidivism_NY_arrest	31,624	20	rearrest for any crime
recidivism_NY_drug	27,526	20	rearrest for drug crime

Table 1: Overview of recidivism prediction datasets considered in Section 4. All datasets are publically available. We include a copy of `compas_violent` and `compas_arrest` with our submission. The remaining datasets must be requested from ICPSR due to privacy restrictions.

We use the raw data from each study to define training datasets by following standard practice in the development of recidivism prediction instruments [see e.g., guidelines for the development of an instrument by the Pennsylvania Sentencing Commission 18]. Each dataset consists of a binary outcome (e.g., a proxy variable such as arrest, arrest for violent crime) and a set of features for prisoners released in a specific state (e.g., NY, CA). We process each dataset by dropping instances with missing entries, binarizing features, and oversampling the minority class to equalize the number of instances with positive and negative labels. Oversampling ensures that we can directly minimize the training error without producing trivial classifiers. This also allows us to report meaningful measures of the error rate as opposed to class-specific error rates. We find that oversampling has a trivial effect on our measures of multiplicity ($< 1\%$). We then split each dataset into a *training set* composed of 80% of points, and a *test set* composed of 20% of the points.

C Supporting Experimental Results

	Baseline Model	Individual Ambiguity Model	Max Discrepancy Model
$h(\mathbf{x}_p)$	+1	-1	-1
Error (Train)	32.7%	32.7%	33.6%
Error (Test)	33.4%	33.4%	34.5%
Discrepancy (Train)	0.0%	0.0037%	16.8%
Discrepancy (Test)	0.0%	0.0%	15.1%
Score Function	+ 0.5 age $\leq$ 25	+ 10.3 age $\leq$ 25	+ 7.7 age $\leq$ 25
	+ 0.0 age $\geq$ 25-45	+ 0.0 age $\geq$ 25-45	+ 0.0 age $\geq$ 25-45
	- 16.4 age $\geq$ 46	- 9.9 age $\geq$ 46	- 7.8 age $\geq$ 46
	- 16.3 female	- 9.7 female	- 7.6 female
	- 0.2 n_priors=0	+ 0.0 n_priors=0	- 7.8 n_priors=0
	- 0.1 n_priors $\geq$ 1	+ 0.0 n_priors $\geq$ 1	+ 0.0 n_priors $\geq$ 1
	+ 16.4 n_priors $\geq$ 2	+ 19.8 n_priors $\geq$ 2	+ 7.4 n_priors $\geq$ 2
	+ 16.6 n_priors $\geq$ 5	+ 10.1 n_priors $\geq$ 5	+ 7.8 n_priors $\geq$ 5
	+ 0.0 n_juvenile_misdemeanors=0	+ 0.0 n_juvenile_misdemeanors=0	+ 0.0 n_juvenile_misdemeanors=0
	- 0.1 n_juvenile_misdemeanors $\geq$ 1	- 0.1 n_juvenile_misdemeanors $\geq$ 1	+ 0.1 n_juvenile_misdemeanors $\geq$ 1
	+ 0.0 n_juvenile_misdemeanors $\geq$ 2	- 10.1 n_juvenile_misdemeanors $\geq$ 2	- 0.1 n_juvenile_misdemeanors $\geq$ 2
	- 32.6 n_juvenile_misdemeanors $\geq$ 5	- 9.5 n_juvenile_misdemeanors $\geq$ 5	- 15.2 n_juvenile_misdemeanors $\geq$ 5
	+ 0.0 n_juvenile_felonies=0	- 9.9 n_juvenile_felonies=0	+ 7.7 n_juvenile_felonies=0
	- 0.2 n_juvenile_felonies $\geq$ 1	- 10.1 n_juvenile_felonies $\geq$ 1	+ 0.0 n_juvenile_felonies $\geq$ 1
	+ 0.3 n_juvenile_felonies $\geq$ 2	+ 0.3 n_juvenile_felonies $\geq$ 2	+ 15.4 n_juvenile_felonies $\geq$ 2
	+ 0.0 n_juvenile_felonies $\geq$ 5	+ 0.0 n_juvenile_felonies $\geq$ 5	+ 0.0 n_juvenile_felonies $\geq$ 5
	- 0.2 charge_degree=M	- 0.2 charge_degree=M	- 7.5 charge_degree=M
	+ 0.0	+ 0.0	- 0.1

Table 2: Competing linear classifiers for `compas_arrest`, and their predictions on a person with features $\mathbf{x}_p$. We show the baseline model that we would deploy (left) and the models trained to evaluate individual ambiguity (middle) and discrepancy (right). Here, the baseline model predicts $h(\mathbf{x}_p) = +1$ while the other models predict $h(\mathbf{x}_p) = -1$. As shown, there exists at least two models with only 1% additional training error that predict that $\mathbf{x}_p$ would not recidivate — one of which makes 16% of predictions differently than the baseline model.

Dataset	n	d	Baseline Error (Train)	Baseline Error (Test)
compas_arrest	5,380	18	32.7%	33.4%
compas_violent	8,768	18	37.7%	37.9%
pretrial_CA_arrest	9,926	22	34.1%	34.4%
pretrial_CA_fta	8,738	22	36.3%	36.3%
pretrial_NY_fta	730	22	25.9%	29.9%
recidivism_CA_arrest	114,522	20	34.4%	34.2%
recidivism_CA_drug	96,664	20	36.3%	36.2%
recidivism_NY_arrest	31,624	20	31.0%	31.8%
recidivism_NY_drug	27,526	20	32.5%	33.6%

Table 3: Performance of baseline classifiers for all datasets. Here, n is the number of instances in the dataset and d is the number of features. We train each model by minimizing the 0-1 loss function on the training data. All 9 models generalize as shown training error is close to test error. This is expected given that we fit models from a simple hypothesis class on datasets where $n \gg d$.

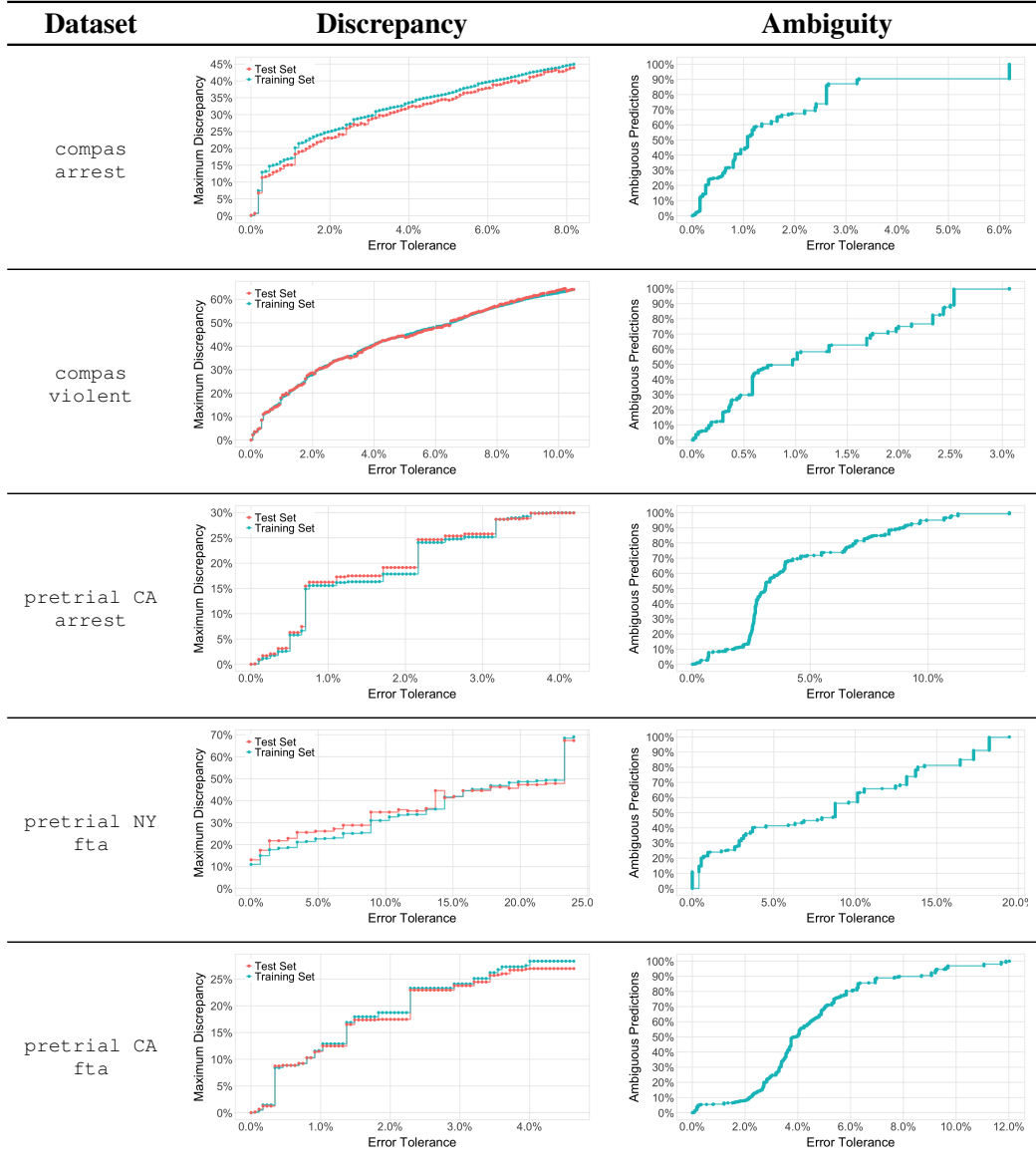


Figure 4: Multiplicity profiles for the `compas` and `pretrial` datasets. The maximum percent of conflicts between the optimal model and any model within the error tolerance (discrepancy) is shown on the left. The percent of individuals who could have been given a different prediction by some model within the error tolerance (ambiguity) is shown on the right.

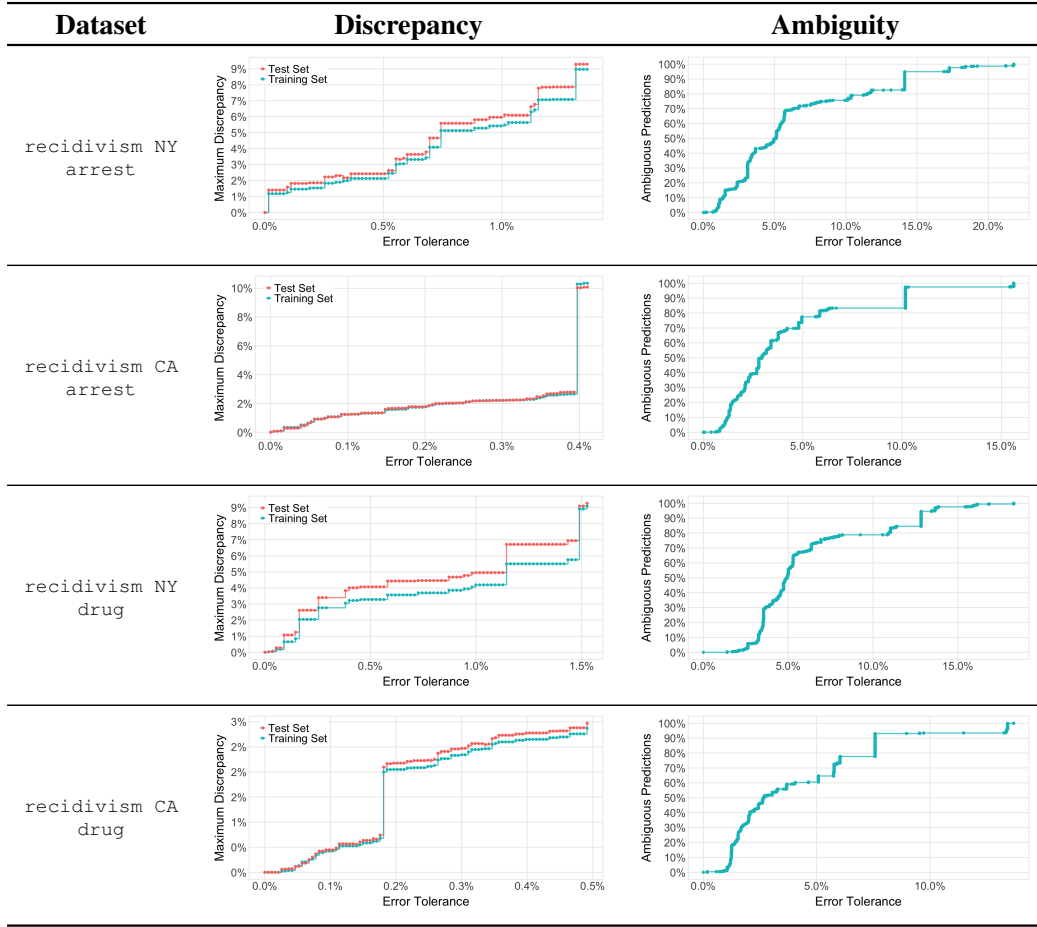


Figure 5: Multiplicity profiles for the `recidivism` datasets. The maximum percent of conflicts between the optimal model and any model within the error tolerance (discrepancy) is shown on the left. The percent of individuals who could have been given a different prediction by some model within the error tolerance (ambiguity) is shown on the right.